Experiment 1: Mixtures

Notes

5. Remove any stirring implement but leave the thermometer in the water and allow it to cool. Be very careful not to touch or bump the beaker or thermometer.

6. Record the temperature at which you observe the formation of any solid.

Procedure

Part C: solubility

1. Test the solubility of a small sample (half a teaspoon) of ethanol, charcoal, sodium chloride and copper(II) chloride in separate 15 mL lots of distilled water. Record your observations and classify each sample as soluble or insoluble in the water solvent column of a table similar to that shown.

Solute	Solubility	
	Water (solvent)	Ethanol (solvent)
charcoal		
sodium chloride		
copper(II) chloride		
ethanol		
water		

2. Test the solubility of a small sample (half a teaspoon) of charcoal, water, sodium chloride and copper(II) chloride in 15 mL lots of ethanol. Record your observations and classify each sample as soluble or insoluble in the ethanol solvent column of your table.

Processing of results and questions

1. Describe how you would make 100 mL of the following types of sodium chloride solution given that the solubility of $NaC\ell$ in water is 35.9 g/100 mL at 25 °C.
   a) unsaturated
   b) saturated
   c) supersaturated

2. Compare temperature records in Part B. Did you manage to make a supersaturated solution? Explain.

3. Which is more soluble in human blood oxygen or carbon dioxide? Record the solubility of the two gasses.

4. Ethanol is the alcohol in alcoholic drinks like wine and beer. Ethanol is said to be miscible in water. What does this mean? Is miscibility different to solubility?